

DECISION-MAKING DRIVEN DESIGN (DMDD): Foundational Architecture for Enterprise Decision Intelligence

A Design Science Working Paper Establishing the Theoretical and Architectural Foundations of the
DMDD Research Program

Dr. Eng. Samir Lotfi Ali

Operations & Digital Transformation Strategist | Author, DMDD Series
Doctorate, Pierre and Marie Curie University (Sorbonne, Paris VI) | Très Honorable

Working Paper | Available at SSRN: [DOI assigned upon upload] | Version: May 2026

Paper 0 of the DMDD Working Paper Series — Series Anchor Document. This working paper establishes the foundational architecture and design science principles of the Decision-Making Driven Design (DMDD) framework. It serves as the primary scholarly reference for all subsequent papers in the series. The complete framework is developed in full technical and practitioner detail in Decision-Making Driven Design (DMDD): Building the Enterprise Decision Twin, Volume 1 (Ali, 2026; Amazon KDP, ASIN: BoGXMGRMJQ). This working paper presents the foundational architecture in design science form for the academic community; it does not reproduce extended passages from the published volume.

Abstract

Enterprise systems have reached a structural inflection point. Organisations now generate more data, deploy more analytical tools, and operate with more interconnected processes than at any prior moment — yet persistently fail to translate computational capability into decision quality, traceability, or institutional learning. The root cause is architectural: enterprise systems are designed to record and execute transactions, not to reason about decisions and compound knowledge over time. This paper establishes the foundational architecture of Decision-Making Driven Design (DMDD) — a design science framework that repositions decision quality, not transaction throughput, as the primary metric of enterprise system performance.

DMDD introduces four foundational constructs: the Enterprise Decision Twin, a living cognitive architecture that operates in parallel with transactional systems; the SDOF Cycle (Scan, Dig, Optimise, Foresee), the perpetual intelligence operating cycle; the Neuro-Intelligent Unified Architecture (NIUA), the seven-layer neural substrate; and the Decision Black Box, the institutional memory architecture that enables compound organisational learning. Together, these constructs constitute an integrated governance architecture for enterprise decision intelligence. This paper positions DMDD within the design science research tradition (Hevner et al., 2004), distinguishes its

contributions from adjacent literatures in data quality, AI governance, and organisational learning, and establishes the research agenda that subsequent papers in the series extend.

Keywords: Decision Intelligence · Enterprise Architecture · Decision-Making Driven Design · DMDD · Enterprise Decision Twin · NIUA · SDOF · Design Science Research · Institutional Memory · AI Governance · Organisational Learning · Human-in-the-Loop · Decision Black Box · Cognitive Architecture

1. The Enterprise Decision Problem: Why Architecture, Not Capability, Is the Gap

The enterprise systems of the past three decades were designed for a world of relative stability. Enterprise Resource Planning (ERP) systems — the dominant infrastructure of enterprise operations since the 1990s — are optimised for transaction processing: capturing, routing, and executing defined processes with consistency and scale. Within the operating conditions for which they were designed, they perform this function with genuine effectiveness.

The operating conditions have changed. Geopolitical instability, technological disruption, supply chain complexity, workforce transformation, and the rapid evolution of AI capabilities have created an environment in which the primary determinant of organisational performance is not the efficiency of transaction execution but the quality of the decisions that govern it. Organisations now require systems that can reason about decisions, validate the inputs that inform them, integrate institutional learning across decision cycles, and govern the human-AI collaboration through which consequential choices are made.

Current enterprise architectures cannot deliver this. The following failure modes — drawn from the foundational analysis in DMDD Volume 1 (Ali, 2026) — are not edge cases. They are structural properties of transaction-centric architecture applied to decision-intelligence problems:

Failure Mode	Architectural Cause
Rigidity Paradox	Systems optimised for standardisation create brittle processes that cannot adapt when operating conditions shift. Customisation accumulates, compounding rigidity over time.
Organisational Amnesia	Transactional systems record what was decided but not why, what alternatives were considered, or what was learned. Institutional knowledge leaves with departing personnel.
Epistemic Rupture	When analytical models fail simultaneously — because the environment has shifted beyond their training conditions — the enterprise has no mechanism to detect model invalidation before consequential decisions compound the error.

Shadow System Proliferation	Employees circumvent official systems through spreadsheets, email, and informal processes when those systems cannot support the operational decisions they face. Intelligence disperses and governance fractures.
Deterministic Error Propagation	Tightly integrated deterministic systems propagate errors across functions with precision. An invalid input cascades into structurally corrupted outputs across multiple downstream processes before detection is possible.
Compound AI Risk	AI deployed atop unchanged enterprise architecture without governed input validation, institutional memory, or structured human oversight becomes an autonomous risk generator: capable, ungoverned, and unable to learn.

Table 1. Enterprise Decision Failure Modes. These failure modes share a common architectural origin: systems designed to record and execute cannot govern, validate, or learn. DMDD addresses this class of failure through a decision-centric architectural redesign.

The aggregate consequence is what DMDD identifies as the enterprise intelligence gap: the structural distance between the volume of data an organisation possesses and the quality of the decisions it can make from it. Closing this gap is not a matter of acquiring more powerful AI models. It is a matter of redesigning the architecture through which intelligence is acquired, validated, governed, and accumulated.

Enterprise systems are accurate. Enterprise decisions are not. The gap between them is architectural, not informational.

1.1 The Research Problem

This paper addresses a precise and tractable research problem: what architectural principles must an enterprise system satisfy in order to produce decisions that are epistemically valid, institutionally accountable, and progressively improving over time?

This is a design science problem (Hevner et al., 2004), not an empirical prediction problem. The contribution is an architectural artefact — the DMDD framework — that specifies the structural conditions under which enterprise decision intelligence can be achieved. Empirical validation of that architecture in deployed contexts is a research objective of subsequent papers in this series.

1.2 Research Methodology

This research adopts a design science approach, following the guidelines of Hevner et al. (2004) for design science research in information systems. The framework is developed as an architectural artefact addressing an identified class of enterprise system failure. The design process is motivated by practitioner-based observation across manufacturing, finance, logistics, and healthcare deployments where the failure modes described in Table 1 were systematically observed. Theoretical integration draws on decision theory (Simon, 1955; Kahneman and Tversky, 1979), organisational memory (Walsh and Ungson, 1991), distributed cognition (Hutchins, 1995), human-machine teaming (Sycara and Lewis, 2004), and data quality research (Wang and Strong, 1996; Pipino et al., 2002).

The architecture is evaluated against the design science criteria of utility, rigour, and relevance. Utility is demonstrated through the architecture’s capacity to address the failure modes identified in Table 1. Rigour is demonstrated through formal specification of each architectural component. Relevance is established through engagement with the research communities in information systems, AI governance, and enterprise architecture to which DMDD contributes.

On Measurement: Decision quality \u2014 the primary construct this architecture is designed to improve \u2014 is operationalised across four measurable dimensions proposed for empirical testing in Paper 4: (1) decision traceability rate, the proportion of consequential decisions for which complete reasoning chains are recoverable; (2) epistemic validity compliance, the proportion of input signals satisfying all VIVO conditions prior to architecture entry; (3) propagation detection latency, time elapsed between introduction of an invalid premise and its detection within synthesis; and (4) institutional memory reuse rate, the frequency with which Decision Black Box precedents are materially applied in subsequent analogous decisions.

2. The DMDD Framework: The Architectural Response

The failure modes identified in Section 1 share a single architectural root cause: enterprise systems were designed to optimise transaction execution, not to govern decision formation. DMDD directly addresses this structural gap.

Decision-Making Driven Design (DMDD) is an enterprise architecture methodology that redesigns enterprise systems around the decision process rather than the transaction process. The fundamental reorientation is from recording what happened to reasoning about what should happen — and from executing instructions to governing choices.

DMDD does not replace ERP systems. It adds the cognitive layer that ERP systems structurally lack: a governed reasoning architecture that operates in parallel with transactional infrastructure, drawing context from it and returning decisions that improve it. The two systems are complementary: ERP handles execution; DMDD handles the governed intelligence that determines what should be executed.

2.1 Five Design Principles

The DMDD framework is grounded in five design principles that distinguish it from both traditional enterprise architecture and conventional AI deployment:

Design Principle	Architectural Consequence
Decision-Centricity	The primary system objective is decision quality, not transaction throughput. Every architectural component is evaluated by its contribution to the validity, traceability, and institutional learning value of decisions.

Pre-Transactional Intelligence	Intelligence must operate upstream of transaction execution, not downstream of it. Governance applied after a decision has been executed addresses consequences; governance applied before addresses causes.
Compound Learning	Each completed decision cycle must enrich the knowledge available to the next. Learning is an architectural property — a structural consequence of how the system records and retrieves — not a cultural outcome dependent on individual behaviour.
Non-Delegable Human Primacy	Human authority over consequential decisions is not a configuration option. It is a structural property of the architecture. The system generates recommendations with full reasoning transparency; humans retain authority over every consequential decision.
Epistemic Validity	A decision is valid when it is formed from inputs that are contextually relevant, scope-complete, and reasoning-processable for the specific decision at hand — not merely when those inputs are data-correct in isolation.

Table 2. DMDD Design Principles and Their Architectural Consequences.

2.2 The Three Rationality Dimensions

DMDD operationalises decision intelligence across three dimensions of organisational rationality, each addressing a distinct failure mode of traditional enterprise systems:

- **Epistemic Rationality — Building Accurate Beliefs:** The capacity to form accurate beliefs about the operational environment from incomplete, conflicting, and continuously changing signals. Traditional systems build beliefs from transactional records — what has already occurred. DMDD builds beliefs from the full signal field, including weak signals that precede transactional events.
- **Computational Rationality — Processing Under Constraint:** The capacity to reason effectively under the constraints of limited time, incomplete data, and bounded human attention. DMDD's architecture does not eliminate these constraints — it manages them structurally through governed prioritisation, parallel processing, and confidence-calibrated human escalation.
- **Normative Rationality — Embedding Ethics and Governance:** The capacity to align decision-making with organisational values, regulatory requirements, and stakeholder interests. DMDD enforces normative rationality through its HITL architecture: human authority is not a policy commitment but a structural property that the system cannot bypass.

These three dimensions are operationalised through the eight-step DMDD rational process, which maps each dimension to specific architectural components across the decision lifecycle. Critically, the three rationality dimensions are not merely theoretical principles — they are the architectural

blueprint for the Enterprise Decision Twin itself: each dimension corresponds directly to one of its three structural layers, as specified in Section 3.

2.3 The Eight-Step Rational Process

The DMDD framework operates through an eight-step rational process that transforms raw enterprise signal into governed, traceable, and institutionally accountable decision intelligence:

Step	Name	Function	Rationality Dimension
1	Pattern Detection	Filters and classifies incoming signals; distinguishes genuine deviation from noise; applies relevance thresholds to focus analytical resources	Epistemic
2	Feasibility Assessment	Evaluates options against regulatory, operational, and governance constraints; eliminates infeasible candidates before analytical resources are committed	Normative
3	Objective Function Definition	Formalises competing stakeholder goals as quantifiable, weighted objective functions; makes strategic trade-offs explicit and auditable	Goal-Oriented
4	Multi-Scenario Generation	Applies context-appropriate analytical methods to generate structured alternative futures; selects scenario architecture based on decision type and uncertainty level	Computational
5	Knowledge Retrieval	Performs case-based reasoning across the Decision Black Box; surfaces analogous historical decisions; synthesises precedent insights for current application	Epistemic
6	Human Synthesis & Validation	Surfaces synthesised intelligence to human decision-makers through adaptive interfaces with full reasoning transparency; HITL governance boundary	Normative
7	Implementation Tracking	Monitors execution against decision intent using rational	Goal-Oriented

		variance analysis; captures ground-truth feedback for model updating	
8	Mass-Scale Learning	Mines decision-outcome patterns across the full institutional history; generates early warning signals; updates model weights and detection parameters	Computational + Epistemic

Table 3. The DMDD Eight-Step Rational Process (Ali, 2026). Each step is both a discrete function and a contribution to the compound learning loop: outputs of Step 8 recalibrate the signal filters of Step 1.

The eight steps do not execute sequentially in a linear pipeline. They operate as a continuously active, perpetually cycling cognitive system: the SDOF Cycle. Each phase of the SDOF Cycle activates specific steps and depends on the outputs of prior phases while contributing to the conditions of subsequent ones.

3. The Enterprise Decision Twin: The DMDD Architecture Instantiated

Having established the three rationality dimensions through which DMDD governs enterprise intelligence (Section 2.2), this section presents their architectural instantiation: the Enterprise Decision Twin. Each rationality dimension becomes a structural layer; the three layers together constitute the complete cognitive architecture.

The Enterprise Decision Twin is the primary architectural construct of DMDD — a living cognitive replica of the organisation’s decision environment that operates in parallel with its transactional systems. The Decision Twin is not a dashboard, a reporting layer, or a predictive model. It is a governed reasoning architecture that continuously models the consequences of decisions before they are made and learns from every decision after it is taken.

The distinction from a digital twin in the engineering sense is important. An engineering digital twin replicates a physical asset for monitoring and simulation. The Enterprise Decision Twin replicates the organisation’s decision-making context — the full epistemic environment in which choices are made, including institutional memory, competing stakeholder objectives, environmental signals, and the reasoning history of prior decisions. It does not simulate the organisation; it reasons about what the organisation should do next.

The Decision Twin is not a tool that processes decisions. It is a cognitive partner that participates in them — preserving the reasoning, learning from the outcomes, and becoming more capable with every cycle.

3.1 Three-Layer Architecture: The Rationality Dimensions as Structural Components

The three rationality dimensions introduced in Section 2.2 are not abstract principles applied to an existing architecture — they are the architecture. Each dimension is implemented as a structural layer of the Enterprise Decision Twin, and each layer is accountable for one dimension of governed intelligence:

Layer	Primary Function	DMDD Role
Layer 1 Epistemic Rationality	The Organisational Knowledge Base: captures the full state of the enterprise — transactional data from ERP, informal contextual knowledge, and the evolving pattern library of organisational learning	Provides the complete epistemic foundation from which the Intelligence Layer reasons; includes the full Decision Black Box history
Layer 2 Computational Rationality	The Intelligence Layer: houses the analytical engines — inference, optimisation, foresight, and real-time critique — that reason across domains, simulate scenarios, evaluate trade-offs, and generate recommendations	The reasoning substrate; governed by NIUA’s seven-layer architecture and driven continuously by the SDOF Cycle
Layer 3 Collaborative Rationality	The Interaction Layer: surfaces intelligence to human decision-makers through adaptive interfaces, causal explanations, and closed-loop execution tracking	The human governance boundary; implements Step 6 of the eight-step process and operationalises HITL as a structural property

Table 4. The Three-Layer Architecture of the Enterprise Decision Twin (Ali, 2026).

The three layers are co-dependent: the Knowledge Base provides the epistemic foundation the Intelligence Layer reasons from; the Intelligence Layer generates the recommendations the Interaction Layer surfaces; the Interaction Layer captures the human decisions and rationale that enrich the Knowledge Base. The system is not a pipeline. It is a governed cognitive loop.

3.2 Key Capabilities

The Decision Twin delivers seven capabilities that conventional enterprise architectures cannot provide simultaneously:

- **Continuous Learning:** Each decision cycle strengthens the system’s analytical foundations. Learning is not a periodic model retraining event; it is a structural property of the architecture.

- **Multi-Objective Trade-off Reasoning:** The system optimises across multiple competing objectives simultaneously, making trade-offs explicit and quantifiable rather than leaving them implicit in human judgement.
- **Weak Signal Detection:** The architecture surfaces low-amplitude early indicators of emerging disruption before they materialise as transactional events — the class of signal that periodic reporting categorically misses.
- **Epistemic Humility:** When learned patterns cease to predict outcomes, the system detects the failure, escalates appropriately, and suspends recommendations resting on invalidated premises.
- **Mandatory Human Primacy:** The Decision Twin generates recommendations with full reasoning transparency. Humans retain authority over every consequential decision. This is enforced architecturally, not stated as policy.
- **Institutional Memory:** The reasoning behind every decision — not merely the decision itself — is permanently retrievable. Institutional knowledge survives personnel change because it resides in the architecture, not in individual memory.
- **Compound Intelligence:** The system that has operated through one thousand decision cycles is not merely more experienced than the one that has operated through one hundred — it is structurally more capable, because its Knowledge Base is richer, its pattern library is deeper, and its model weights are more precisely calibrated.

3.3 The Architecture in Action: A Hypothetical Vignette

The following structured vignette illustrates how the four DMDD constructs operate as an integrated system in a realistic enterprise scenario. It is not an empirical case study; it is a theoretical demonstration of architectural logic.

Scenario: A global manufacturing group faces an unexpected raw material supply disruption. A key supplier has halted shipments following a geopolitical development. Existing ERP systems flag the inventory shortage but provide no governed reasoning about what to decide or why.

- **SCAN (Decision Twin, Layer 1–2):** Six weeks prior to the formal shortage alert, the Cognitive Sensing Layer detected a cluster of weak signals — changes in supplier communications, logistics route anomalies, and a geopolitical development in the supplier's region. These signals passed the relevance threshold and entered the architecture. The Decision Black Box simultaneously surfaced three analogous historical episodes, including one that resulted in a nine-month production halt.
- **DIG (Layer 2, Decision Black Box retrieval):** The DIG attractor retrieved the full decision traces from the analogous historical episodes. The causal chains were reconstructed: in two of three cases, delayed diversification response had compounded the disruption. The institutional memory contained a specific lesson: alternative supplier qualification timelines had been consistently underestimated in the organisation's prior planning.
- **OPTIMISE (Layer 3–4, HITL synthesis):** The Neuro-Decision Core generated three ranked options with explicit assumptions and confidence scores. Before synthesis was finalised, the HITL component at Layer 5 was activated: the decision-maker identified that the top-ranked option assumed a 45-day alternative supplier qualification timeline that relationship intelligence indicated was unavailable. Option A was set aside. Option B was selected with a modification, captured with full rationale in the Decision Black Box.

- **FORESEE and Layer 7 Learning:** The decision cycle — including the HITL override, its rationale, and the rejected option — was recorded in full. The FORESEE attractor generated anticipatory queries for the next SCAN cycle: monitor secondary supplier geopolitical exposure, track qualification timelines for currently approved alternates, flag any recurrence of the weak signal cluster that preceded this event. The next analogous decision begins from a structurally richer context.

This vignette demonstrates five architectural properties operating simultaneously: pre-transactional signal detection, institutional memory retrieval, human epistemic authority embedded in synthesis, full decision traceability, and compound learning that enriches the next cycle. No component operates in isolation; the system’s value emerges from their governed integration.

4. The SDOF Cycle: Perpetual Intelligence Operation

The SDOF Cycle — Scan, Dig, Optimise, Foresee — is the operating logic of the Enterprise Decision Twin. It is not a sequential workflow, a project lifecycle, or a planning cycle. It is a perpetual cognitive spiral: four attractors in continuous circulation, each phase enriching the conditions of the next, each completed rotation strengthening the analytical foundations of subsequent rotations.

The term “attractor” is chosen deliberately. In dynamical systems theory, an attractor is a state toward which a system continuously evolves. Each SDOF phase is a cognitive attractor: the system is continuously drawn toward sensing (SCAN), toward understanding (DIG), toward choosing (OPTIMISE), and toward anticipating (FORESEE). There is no resting state; there is no completion. The cycle is the system.

SDOF is not a linear workflow. It is a perpetual cognitive system where each cycle reshapes the next through accumulated organisational learning. The system that enters SCAN for the thousandth time does so from a fundamentally richer epistemic foundation than the one that entered it for the first.

The spiral is not circular repetition. It is an upward, compounding progression: each completed cycle deposits a full decision trace into the Decision Black Box, which the next cycle retrieves during its DIG phase. The cognitive substrate available to the system after one thousand cycles is categorically richer than what was available after ten — not merely because more data exists, but because more governed reasoning has been structured and made retrievable.

Attractor	Core Function	Key Mechanism	Table 3 Steps
SCAN	Perpetual environmental scanning: detects signals across all domains before they appear in transactional records	Multi-dimensional signal field; weak signal amplification; relevance threshold filtering	Steps 1 & 3 (see Table 3)

DIG	Historical intelligence extraction: retrieves and reconstructs patterns from the Decision Black Box in response to escalated signals	Case-based reasoning; pattern matching; temporal sequencing; institutional memory retrieval	Steps 2 & 5 (see Table 3)
OPTIMISE	Multi-objective decision synthesis: generates and evaluates decision options across competing stakeholder objectives under current constraints	Scenario architecture; multi-objective optimisation; causal modelling; confidence scoring	Steps 3, 4 & 6 (see Table 3)
FORESEE	Anticipatory intelligence generation: constructs structured plausible futures and feeds them forward as pre-positioning queries for the next SCAN cycle	Scenario propagation; distributional outcome modelling; FORESEE-to-SCAN anticipatory loop	Steps 7 & 8 (see Table 3)

Table 5. The SDOF Four-Attractor Cycle (Ali, 2026). The FORESEE-to-SCAN anticipatory loop is the mechanism through which the cycle compounds: each FORESEE output generates pre-positioned queries that accelerate the next SCAN, progressively reducing the latency between emerging signal and governed decision.

4.1 The Compounding Mechanism

The SDOF Cycle produces compound intelligence through a specific architectural mechanism: each completed cycle deposits outputs in the Decision Black Box (Layer 7 of NIUA), and the next cycle draws from that enriched deposit in its DIG phase. The result is not linear accumulation — each new deposit interacts with prior deposits through pattern recognition, producing insights that neither alone could generate.

Practically: an organisation operating the SDOF Cycle for three years has not simply accumulated three years of data. It has accumulated three years of governed decision traces — each including the signal context, the options considered and rejected with their rationale, the assumptions made, and the outcome observed against the prediction. The analytical depth available to the third year’s OPTIMISE phase is categorically different from what was available in the first.

5. NIUA: The Seven-Layer Neural Substrate

The Neuro-Intelligent Unified Architecture (NIUA) is the computational substrate of the Enterprise Decision Twin — the seven-layer neural architecture that implements the DMDD eight-step process

across the Decision Twin’s three layers. NIUA is not AI-enabled ERP. It is cognitive enterprise architecture designed for intelligence from the ground up.

The distinction is architectural, not rhetorical. Traditional ERP processes transactions sequentially, in batches, through pre-programmed logic. NIUA processes information flows continuously, in parallel, and adaptively — reasoning about what signals mean and what should happen next rather than executing what has already been decided. Where ERP isolates functional modules, NIUA creates networked intelligence: each layer communicates with adjacent layers the way neurons communicate in a cognitive system, not the way database tables communicate through SQL joins.

Terminology Note: The term 'neural' in NIUA refers to the architecture’s topology and operating logic — distributed, parallel, adaptive connections between layers — rather than implying that every layer is implemented as a deep learning neural network. Some layers (notably Layers 1, 6, and 7) may be implemented using rule-based engines, structured databases, or symbolic AI components. The neural metaphor describes how layers interact and adapt, not the computational substrate of any specific layer.

L	Name	Function	Design Science Role
1	Data Source Neurons	Ingests real-time signals from all enterprise sources across modalities (structured, unstructured, time-series, and weak signals); implements relevance threshold filtering aligned with current decision context	Epistemic input layer; feeds Step 1 (Pattern Detection)
2	Unified Data Lake	Provides persistent storage with full temporal tracking; integrates current signals with complete Decision Black Box history; enables time-travel retrieval of prior decision contexts	Organisational Knowledge Base; feeds Step 5 (Knowledge Retrieval)
3	Feature Engineering	Implements multi-head attention for contextual pattern detection; transforms raw signals into decision-relevant analytical features; identifies non-obvious correlations below observable data surfaces	Pattern recognition substrate; feeds Steps 2, 3
4	Neuro-Decision Core	Primary reasoning layer: synthesises analytical outputs across all domains, generates ranked decision options with explicit reasoning traces, and implements the Proactive Challenge Mechanism for epistemic rupture detection	Intelligence synthesis; feeds Steps 3, 4, 6

5	Human-in-the-Loop Integration	The non-delegable governance boundary: operationalises three collaboration modes (Human-as-Operator, Human-as-Neuron, Human-as-Trainer) calibrated to decision confidence and stakes; manages HITL escalation protocol	Governance layer; implements Step 6; non-bypassable for consequential decisions
6	Action Execution	Translates governed decisions into specific system commands and workflow triggers with full provenance; monitors implementation against decision intent	Execution layer; implements Step 7
7	Outcome Learning	Captures decision-outcome pairs with complete context; updates model weights; implements mass-scale pattern mining; feeds compound intelligence back to Layer 2 for the next cycle	Institutional memory; implements Step 8; feedback to Layer 1

Table 6. The NIUA Seven-Layer Architecture (Ali, 2026). The feedback connection from Layer 7 (Outcome Learning) to Layer 1 (Data Source Neurons) is the architectural mechanism of compound intelligence: each completed decision cycle recalibrates the signal filters that govern the next.

5.1 Three Operational Modes

Layer 5 of NIUA — the Human-in-the-Loop Integration layer — operationalises three collaboration modes, each appropriate for different combinations of decision confidence and stakes. Mode selection is automatic, based on the system’s confidence score for the current decision context:

- **Mode 1 — Human as Operator (confidence < 0.70):** Human makes all decisions; AI provides analysis only. Applied to novel situations, high-stakes strategic decisions, and early deployment contexts where trust calibration is incomplete.
- **Mode 2 — Human as Neuron (0.70 ≤ confidence < 0.90):** Human and AI collaborate; expert human judgement becomes a high-weight input signal ($w = 0.90$) to the analytical synthesis rather than a post-hoc review. This is the default operating mode for the majority of enterprise decisions.
- **Mode 3 — Human as Trainer (confidence ≥ 0.90):** AI makes autonomous decisions within defined boundaries; human monitors exceptions and provides meta-feedback for model evolution. Applied only to routine decisions with a proven track record of model reliability.

The three modes represent a spectrum of human-AI collaboration calibrated to the actual epistemic conditions of each decision. The architecture does not default to maximum autonomy — it defaults to maximum governance, escalating to autonomy only as demonstrated reliability justifies it.

Calibration Note: The confidence thresholds of 0.70 and 0.90 presented here are illustrative reference points, not universal fixed values. Actual deployment should calibrate these thresholds to the organisation's specific risk tolerance, regulatory environment, and demonstrated model reliability in the decision domain concerned. In high-stakes or regulated contexts (healthcare, financial services, critical infrastructure), Mode 3 autonomous operation may be architecturally excluded regardless of confidence score. The thresholds are configurable design parameters, not architectural constants.

5.2 Epistemic Rupture Detection

A distinctive capability of the NIUA architecture is its structured response to epistemic rupture — the condition in which the analytical models that the system uses to interpret signals have been invalidated by environmental shift. This is the failure mode that conventional AI systems cannot detect from the inside: a model that has been trained on pre-disruption data will continue generating confident-sounding predictions after the disruption has made those predictions structurally invalid.

NIUA addresses this through the Proactive Challenge Mechanism at Layer 4: a continuously active Reasoning Integrity Vector that compares predicted outcomes against actual outcomes across deduction, induction, and abduction dimensions. When the decay rate of predictive accuracy crosses defined thresholds, the system escalates to Layer 5 (HITL), suspends recommendations resting on the invalidated premises, and initiates a DIG re-query to reconstruct the analytical foundation from currently valid signals.

Epistemic rupture is not a model failure. It is an environmental event that the model cannot detect without a dedicated detection architecture. NIUA builds that architecture in at Layer 4.

6. The Decision Black Box: Institutional Memory Architecture

A Note on Terminology: The term 'Decision Black Box' is a deliberate inversion of its conventional meaning. In aviation and engineering, a black box is passive, opaque, and accessed only after failure. The DMDD Decision Black Box is its architectural opposite: it is active, transparent, and preventive — continuously queried before and during decision cycles, not forensically examined after them. The name is retained precisely because that inversion captures something important: the institutional knowledge that most organisations can only retrieve after failure is, in DMDD, the primary reasoning resource accessed before every consequential decision.

The Decision Black Box is the institutional memory layer of the Enterprise Decision Twin — Layer 7 of NIUA and the architectural mechanism through which DMDD converts individual decision cycles into compounding organisational intelligence. The fundamental problem it addresses is what DMDD calls organisational amnesia: the structural loss of decision reasoning through personnel change, system limitations, and the absence of mechanisms for capturing why decisions were made, not merely what was decided.

The fundamental problem the Decision Black Box addresses is what DMDD calls **organisational amnesia**: the structural loss of decision reasoning through personnel change, system limitations, and the absence of mechanisms for capturing why decisions were made, not merely what was decided. Every organisation that has watched an experienced leader depart, taking with them the contextual

reasoning behind years of consequential choices, has experienced organisational amnesia. The Decision Black Box resolves it architecturally.

Traditional systems store what was decided. The Decision Black Box stores why it was decided, what alternatives were available, what was predicted, and what actually happened. That distinction is the difference between a historical record and an institutional intelligence base.

What the Decision Black Box Captures	Governance Function
Decision context and environment	Reconstructs the full epistemic situation at decision time; enables retrospective evaluation of whether the decision was valid given available information
Alternatives considered and rationales for rejection	Makes the decision's option space retrievable; surfaces options that were available but not taken as analytical inputs for analogous future decisions
Predictions and expectations at decision time	Enables comparison of predicted outcomes against actual outcomes; the foundation of model calibration and early warning system development
Actual outcomes with comparative analysis	Closes the learning loop; feeds Steps 7 and 8 of the eight-step process; recalibrates model weights and detection thresholds
Informal context and communications	Captures the tacit knowledge that shaped the reasoning but does not appear in structured data; the element that departing personnel would otherwise take with them
Human contributions and overrides	Records the specific contextual judgements through which human authority modified or overrode AI recommendations; encodes human expertise as retrievable institutional knowledge

Table 7. Decision Black Box Capture Dimensions and Governance Functions (Ali, 2026).

6.1 Compound Learning Mechanism

The compound learning mechanism of the Decision Black Box operates through a specific architectural sequence. Each completed decision cycle deposits a full decision trace into the Black Box at Layer 7. The next cycle's DIG phase retrieves relevant precedents from this enriched deposit. The OPTIMISE phase synthesises current analysis with retrieved precedent patterns. And the FORESEE phase generates forward projections that are pre-calibrated against the compound pattern library.

The compound advantage accumulates across four dimensions: reducing repeat failures by making the reasoning behind past mistakes retrievable; accelerating organisational learning by making lessons available across the full organisation in days rather than months; enabling proactive risk

detection by identifying systemic risk patterns before they cascade; and preserving institutional intelligence through personnel change because knowledge resides in the architecture, not in individual memory.

6.2 Governance Architecture

The Decision Black Box is designed for organisational learning, not individual accountability. Decision records are structured to support process improvement analysis; the governing question the system asks is always what the organisation can learn from this cycle, not who is at fault for a given outcome. This distinction is architecturally enforced through anonymisation of individual decision-makers in analytical outputs, ensuring that the intelligence generated from the system’s history is directed toward future improvement rather than retrospective attribution.

7. Design Science Positioning and Research Contributions

DMDD addresses an identified and persistent class of enterprise system failure through a new architectural artefact. Its design science credentials rest on three properties: the artefact addresses a real and well-documented problem (the enterprise intelligence gap); it proposes a rigorous solution that cannot be achieved by the combination of existing artefacts; and it is evaluated against the criteria of utility, rigour, and relevance.

7.1 Distinction from Adjacent Literatures

DMDD’s contributions are distinct from, though informed by, several adjacent research fields:

Adjacent Field	What It Addresses	DMDD’s Distinct Contribution
Data Quality Research (Wang & Strong, 1996; Pipino et al., 2002)	Correctness, completeness, and consistency of data as attributes of the data itself	Decision validity: whether data is contextually relevant, scope-complete, and reasoning-processable for the specific decision at hand. A formally distinct construct requiring a new governance architecture.
AI Governance Frameworks (OECD, 2019; NIST, 2023; EU AI Act, 2024)	Model transparency, bias mitigation, accountability, and regulatory compliance for AI outputs	Architectural governance of the full decision lifecycle: from pre-transactional signal acquisition through validated reasoning, governed execution, and institutional learning. Governance applied before and within the decision process, not only at the output boundary.
Organisational Memory (Walsh & Ungson, 1991)	Conceptual frameworks for how organisations store and retrieve	The Decision Black Box as an architectural implementation of institutional memory: structured, automatically populated, continuously

	knowledge through individuals, culture, and systems	retrieved, and compound-learning-enabled. Operationalises the conceptual framework as a technical architecture.
Distributed Cognition (Hutchins, 1995)	How cognitive processes are distributed across individuals, artefacts, and environments in complex systems	The Enterprise Decision Twin as a governed distributed cognitive architecture: formalises the distribution of decision cognition across human agents and AI layers, specifying the governance mechanisms that maintain coherence and accountability.
Decision Intelligence (Pratt, 2019; commercial platforms)	Structured decision processes augmented by AI; includes decision modelling, outcome tracking, and analytical layer integration	DMDD extends Decision Intelligence by adding three properties it does not systematically provide: (1) the Decision Black Box as a compound institutional memory architecture that learns across cycles, not merely within them; (2) HITL embedded within synthesis rather than appended at the output boundary; and (3) the SDOF cycle as a perpetual operating logic rather than a project-phase model. DMDD's governance architecture operates pre-transactionally; most DI frameworks operate post-analytically.
Enterprise Architecture (ERP, BPM, BPMN)	Process execution governance, workflow management, and system integration for operational consistency	Decision formation governance: the architectural layer that determines what should be executed, not only how execution proceeds. DMDD operates upstream of process execution, not as a substitute for it.

Table 8. DMDD's Design Science Positioning Relative to Adjacent Literatures.

7.2 Architectural Responses to Identified Failure Modes

Design science requires that an artefact demonstrably address the problems that motivated its design. The following table closes the loop between the enterprise failure modes identified in Table 1 and the specific architectural components through which DMDD resolves each:

Failure Mode (Table 1)	DMDD Architectural Resolution	Primary Component
Rigidity Paradox	Decision-centric design replaces static rule execution with adaptive reasoning	SDOF Cycle — continuous attractor logic overrides deterministic batch processing

	governed by current decision context, not prior configuration	
Organisational Amnesia	Every decision trace — context, options considered, rationale, prediction, outcome — is permanently structured and retrievable	Decision Black Box (Layer 7 NIUA) — DIG phase retrieval in every cycle
Epistemic Rupture	Proactive Challenge Mechanism at Layer 4 detects model invalidation through Reasoning Integrity Vector; escalates to HITL; suspends invalid recommendations	NIUA Layer 4 — Proactive Challenge Mechanism + Layer 5 HITL escalation protocol
Shadow System Proliferation	Decision Twin operates as the governed intelligence layer that ERP cannot provide, eliminating the operational need that drives shadow system adoption	Enterprise Decision Twin — Layer 3 (Collaborative Rationality) as the sanctioned intelligence interface
Deterministic Error Propagation	VIVO validation gate prevents invalid signals from entering the reasoning architecture; HITL synthesis governance detects compound errors before consequential execution	NIUA Layer 1 signal filtering + Layer 4 synthesis governance + Layer 5 HITL
Compound AI Risk	DMDD governs the full AI decision lifecycle: input validation, human synthesis authority, execution traceability, and institutional learning — eliminating ungoverned autonomy as a structural possibility	Full NIUA stack — all seven layers operating as integrated governance architecture

Table 8b. Mapping of Enterprise Failure Modes to DMDD Architectural Resolutions.

7.3 Primary Scholarly Contributions of This Paper

This foundational working paper makes the following primary scholarly contributions:

- **The DMDD Framework as a Design Science Artefact:** A formally specified enterprise architecture methodology addressing decision quality, traceability, and institutional learning as primary architectural objectives.
- **The Enterprise Decision Twin Concept:** A living cognitive architecture operating in parallel with transactional systems, formally distinct from both ERP and conventional AI deployment.
- **The SDOF Cycle as a Perpetual Cognitive Operating Model:** A non-linear, continuously active intelligence cycle that produces compound organisational learning through its structural coupling with the Decision Black Box.

- **The NIUA Seven-Layer Specification:** A neural substrate architecture implementing decision governance across perception, contextualisation, reasoning, human oversight, execution, and compound learning.
- **The Decision Black Box as Institutional Memory Architecture:** An architectural implementation of organisational memory that makes institutional intelligence compound over time as a structural property, not a cultural outcome.

7.4 Research Agenda

This working paper establishes the foundational architecture. The DMDD Working Paper Series develops the framework across ten papers organised into four phases: conceptual foundations and historical context (Papers 0\20130.5), architecture specification and application (Papers 1\20134), construct formalisation and governance application (Papers 5\20138), and empirical validation and adoption pathway (Papers 9\201310). Each paper is accompanied by a practitioner-facing LinkedIn article that translates the scholarly contribution for a broader professional audience.

Paper	Title	Research Focus
0	DMDD: Foundational Architecture for Enterprise Decision Intelligence (this paper)	What architectural principles must enterprise systems satisfy to produce valid, traceable, and progressively improving decisions?
0.5	Why Industrial Revolutions Keep Accelerating: The Historical Imperative for Decision-Centric Architecture	Why is the transition from transaction-centric to decision-centric architecture the defining architectural inflection point of this technological era?
1	Decision Intelligence in the Age of Agentic AI: Governing Kimi K2.6 Within the DMDD Cognitive Architecture	How does the DMDD architecture govern distributed multi-agent AI systems at enterprise scale, and what failure modes does it resolve?
2	Eliminating the Data Gathering Blind Spots: The Panorama 360\2013Cognitive Sensing Layer	How does governed pre-transactional signal acquisition eliminate the sensing blind spots through which enterprise intelligence failures originate?
3	The Gravitropic Enterprise: Engineered Institutional Memory as the Architectural Foundation of Decision Resilience	How does the Decision Black Box architecture enable organisational resilience, institutional regeneration after disruption, and compound learning across leadership transitions?
4	The Information Black Hole: When Organisations Destroy Their Own Decision Intelligence	What are the structural mechanisms through which enterprise systems generate, suppress, and lose decision-relevant intelligence \2014 and how does DMDD resolve each?

5	VIVO: Valid In, Valid Out \u2014 A Formal Framework for Enterprise Decision Validity	How does the VIVO principle formally distinguish decision validity from data correctness, and what governance architecture does that distinction require?
6	The Rigidity\u2013Agility Trap: Adaptive Decision Governance as the Architectural Resolution	How does DMDD's adaptive governance architecture resolve the structural tension between standardisation and agility that conventional enterprise systems cannot escape?
7	Human and AI Imperfection: Governing Fallible Intelligence in Enterprise Systems	How does an enterprise architecture designed for compound intelligence govern the compounding interaction of human cognitive limitations and AI error modes?
8	The Break-Even Point of Survival: Decision Architecture and Enterprise Resilience Thresholds	At what point does the absence of a decision governance architecture become existentially terminal \u2014 and what structural conditions define that threshold?
9	Empirical Validation of DMDD: Case-Based Research Across Enterprise Deployments	Does the DMDD architecture produce measurable improvements in decision quality, traceability, and institutional learning in enterprise contexts?
10	DMDD Maturity Model: Adoption Pathways and Governance Staging	How do organisations at different governance maturity levels implement, stage, and validate DMDD adoption across the full architecture?

Table 9. The DMDD Working Paper Series Research Agenda \u2014 Ten Papers. Papers 0\u20134 are complete or near completion. Papers 5\u20138 are in development. Papers 9\u201310 are planned for the empirical and maturity validation phase. LinkedIn companion articles accompany Papers 1\u20134 as practitioner-facing public outreach, and are distinct from the scholarly working paper series.

8. Scope, Limitations, and Future Research

Scope Statement: This paper presents the foundational architecture of DMDD as a design science artefact. It does not provide empirical validation of the architecture in deployed enterprise contexts. The architectural claims specify the structural conditions under which enterprise decision intelligence can be achieved; they do not demonstrate that any specific deployment has achieved it. Empirical validation through case study, action research, and implementation-based research is the objective of Paper 4 in this series. Readers should interpret the contributions as architectural and theoretical, with empirical demonstration as the stated next phase of the research program.

Three specific limitations bound the current contribution and define the research frontier:

- **Implementation Specification Gap:** This paper specifies the architecture at the design science level. The technical implementation specifications — algorithm selection, technology stack options, integration protocols with specific ERP systems — are developed in Volume 1 (Ali, 2026) and are not reproduced here. Readers requiring implementation detail are directed to that source.
- **Maturity and Adoption Pathway Gap:** The architecture as specified represents a target state of full implementation. Organisations at different governance maturity levels will require staged adoption pathways. A DMDD Maturity Model is planned as Paper 5 of this series.
- **Empirical Validation Gap:** The architecture's performance claims — improved decision traceability, reduced propagation of epistemic rupture, compound improvement in decision quality over time — are theoretically grounded and architecturally specified but not yet empirically measured. This gap is the primary research objective of Paper 4.
- **Legacy Integration and Brownfield Deployment:** The architecture as formalised does not require replacement of existing ERP infrastructure. The Decision Twin operates as a governed intelligence layer in parallel with transactional systems, drawing from ERP data at Layer 1 while adding cognitive capabilities that ERP cannot provide. However, legacy data quality and master data management standards directly affect the epistemic validity of Layer 1 inputs. Organisations with significant data quality deficits will encounter a preparatory phase before full NIUA deployment is viable. This 'brownfield' adoption challenge — layering governed intelligence over imperfect operational reality — is recognised as a fourth implementation boundary, addressed in part through the VIVO validation gate (Valid In, Valid Out — the epistemic validity gate that governs signal admission to the architecture) and planned further in the Maturity Model of Paper 5.

8.5 Boundary Conditions: Where DMDD May Not Be Appropriate

A rigorous architectural framework must specify not only where it applies but where it does not. DMDD is designed for a specific class of enterprise decision problem; the following conditions indicate contexts where adoption may not be the appropriate or prioritised intervention:

- **Highly Stable Operational Environments:** Organisations in conditions of low environmental volatility, stable competitive landscapes, and predictable demand derive limited marginal value from cognitive sensing and compound learning. Existing ERP infrastructure may be sufficient.
- **Low Decision Complexity and Volume:** Businesses whose decision landscape consists primarily of routine, rule-governed operational choices without strategic uncertainty, multi-stakeholder trade-offs, or weak signal exposure do not require the full NIUA stack.
- **Insufficient Governance Maturity:** Organisations without established data governance, master data management standards, or a culture of decision accountability will encounter structural difficulties at every layer. DMDD amplifies governance quality; it cannot substitute for its absence.
- **Latency-Critical Operations:** Contexts where decision speed is the overriding constraint — high-frequency trading, real-time process control, emergency response — are not appropriate for governed synthesis. The HITL boundary introduces deliberative latency incompatible with millisecond decision requirements.
- **Early-Stage Organisations Without Institutional History:** The Decision Black Box's compound learning value accumulates over time. Organisations in early

operational stages, without the decision history the DIG attractor retrieves, will experience the architecture as overhead rather than leverage.

These boundary conditions are not failures of the framework \u2014 they are the honest specification of the problem class for which it was designed. DMDD delivers maximum value in organisations facing high environmental complexity, multi-stakeholder decision trade-offs, significant institutional memory fragility, and enterprise-scale AI governance demands.

9. Conclusion

Enterprise decision intelligence is not a problem that more powerful AI models will resolve. The organisations that deploy the most capable available model within an ungoverned architecture will compound their existing failure modes at greater speed and scale. The problem is architectural: the absence of a governed structure through which signals are validated, decisions are formed with institutional accountability, and learning accumulates as compound organisational intelligence.

Decision-Making Driven Design addresses this through four integrated architectural constructs: the Enterprise Decision Twin as the governed cognitive partner to transactional systems; the SDOF Cycle as the perpetual operating logic that drives continuous intelligence; the NIUA architecture as the seven-layer neural substrate that implements the full eight-step rational process; and the Decision Black Box as the institutional memory architecture that converts every decision cycle into the foundation of the next.

Together, these constructs define an architectural program: not a tool to deploy, but a governance philosophy to implement. The ambition of the DMDD research series is to establish that program as a rigorous, empirically validated, and practically adoptable framework for the organisations that will navigate the next decade of enterprise AI.

ERP records transactions. AI generates outputs. DMDD governs decisions. The organisations best positioned for the next decade of enterprise AI may be those that understand this distinction — and build the governance architecture that makes it operational.

Appendix A. Key Acronyms and Definitions

The following table provides brief definitions for the primary acronyms used in this paper. Extended formal definitions appear within the relevant sections and in DMDD Volume 1 (Ali, 2026).

Acronym	Full Form	Brief Definition
---------	-----------	------------------

DMDD	Decision-Making Driven Design	The enterprise architecture methodology that positions decision quality as the primary system objective
NIUA	Neuro-Intelligent Unified Architecture	The seven-layer neural substrate that implements the DMDD eight-step process across the Enterprise Decision Twin
SDOF	Scan-Dig-Optimise-Foresee	The four-attractor perpetual intelligence operating cycle of the Enterprise Decision Twin
VIVO	Valid In, Valid Out	The epistemic validity gate governing signal admission to the NIUA architecture; requires contextual relevance, scope completeness, and reasoning processability
HITL	Human-in-the-Loop	The non-delegable governance boundary at NIUA Layer 5; human authority over consequential decisions is enforced architecturally, not by policy
DBB	Decision Black Box	The institutional memory layer (NIUA Layer 7); records complete decision traces for compound learning and governance traceability
DSR	Design Science Research	The research methodology adopted in this paper; artefact-centred inquiry evaluated against utility, rigour, and relevance criteria (Hevner et al., 2004)
USIS	Unified Symbiotic Intelligence Science	The broader theoretical framework under development; anticipates DMDD's governance principles in a scientific theory of distributed human-machine intelligence (Ali, manuscript under review)

References

- Ali, S. L.** (2026). Decision-Making Driven Design (DMDD): Building the Enterprise Decision Twin. Volume 1: The Foundation. Amazon KDP. ASIN: BoGXMGRMJQ. <https://a.co/d/ofawIefY>
- Ali, S. L.** (2026, forthcoming). Decision Intelligence in the Age of Agentic AI: Governing Kimi K2.6 Within the DMDD Cognitive Architecture. DMDD Working Paper Series, Paper 1. The Symbiotic Intelligence Library. [SSRN preprint; posted concurrently with this paper.]
- Ali, S. L.** (2026, forthcoming). The Gravitropic Enterprise: Engineered Institutional Memory as the Architectural Foundation of Decision Resilience. DMDD Working Paper Series, Paper 3. The Symbiotic Intelligence Library. [Working paper in preparation.]
- Ali, S. L.** (manuscript under review). Unified Symbiotic Intelligence Science (USIS). Submitted to MIT Press.

- Hevner, A. R., March, S. T., Park, J., & Ram, S.** (2004). Design science in information systems research. *MIS Quarterly*, 28(1), 75–105.
- Hollnagel, E., & Woods, D. D.** (2005). *Joint cognitive systems: Foundations of cognitive systems engineering*. CRC Press.
- Hutchins, E.** (1995). *Cognition in the wild*. MIT Press.
- Kahneman, D., & Tversky, A.** (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291.
- Levinthal, D. A., & March, J. G.** (1993). The myopia of learning. *Strategic Management Journal*, 14(S2), 95–112.
- Pipino, L. L., Lee, Y. W., & Wang, R. Y.** (2002). Data quality assessment. *Communications of the ACM*, 45(4), 211–218.
- Simon, H. A.** (1955). A behavioral model of rational choice. *The Quarterly Journal of Economics*, 69(1), 99–118.
- Sycara, K., & Lewis, M.** (2004). Integrating intelligent agents into human teams. In E. Salas & S. M. Fiore (Eds.), *Team cognition: Understanding the factors that drive process and performance* (pp. 203–231). American Psychological Association.
- Walsh, J. P., & Ungson, G. R.** (1991). Organizational memory. *Academy of Management Review*, 16(1), 57–91.
- European Parliament.** (2024). Regulation (EU) 2024/1689 on Artificial Intelligence (EU AI Act). Official Journal of the European Union.
- National Institute of Standards and Technology (NIST).** (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). U.S. Department of Commerce. <https://doi.org/10.6028/NIST.AI.100-1>
- OECD.** (2019). Recommendation of the Council on Artificial Intelligence. OECD/LEGAL/0449. Organisation for Economic Co-operation and Development.
- Wang, R. Y., & Strong, D. M.** (1996). Beyond accuracy: What data quality means to data consumers. *Journal of Management Information Systems*, 12(4), 5–33.

About the Author

Dr. Eng. Samir Lotfi Ali is an Operations and Digital Transformation Strategist with over two decades of implementation experience across manufacturing, finance, healthcare, and logistics spanning the Middle East, Europe, Asia, and North America. He holds a Doctorate in Organisation Management from Pierre and Marie Curie University (Sorbonne, Paris VI), awarded Très Honorable. He is the author of *Decision-Making Driven Design (DMDD): Building the Enterprise Decision Twin* (Volume 1, 2026) and the lead researcher of the Unified Symbiotic Intelligence Science (USIS) framework, submitted to MIT Press. The DMDD series is published as part of The Symbiotic Intelligence Library. | [linkedin.com/in/dr-eng-samir-lotfi-ali-890727211](https://www.linkedin.com/in/dr-eng-samir-lotfi-ali-890727211) | drsamirlotfiali@gmail.com

© 2026 Dr. Eng. Samir Lotfi Ali. This working paper is released for scholarly comment and discussion. The frameworks described herein (DMDD, NIUA, SDOF, Decision Black Box, Enterprise Decision Twin) are proposed architectural constructs. For correspondence regarding formal standards development, contact the author.